Real Numbers & Algebraic Expressions

A Chapter 1 practice worksheet — classifying, comparing, and working with real numbers, plus writing and evaluating expressions.

Name: _____ **Grade:** _____

Date: _____ **Weekday:** _____

Time: _____

1. A. Classifying Real Numbers

Every real number belongs to at least one of these families. Some numbers belong to several.

The Real Number Family

Natural numbers $(1, 2, 3, \dots)$ are the counting numbers.
Whole numbers $(0, 1, 2, 3, \dots)$ are the natural numbers plus 0.
Integers $(\dots, -2, -1, 0, 1, 2, \dots)$ are the whole numbers plus their opposites.
Rational numbers can be written as a fraction $\frac{a}{b}$ of two integers ($b \neq 0$) — this includes all integers, all terminating decimals, and all repeating decimals.
Irrational numbers *cannot* be written as a fraction of two integers — their decimals go on forever with no repeating pattern (like $\sqrt{2}$ or π). Every real number is rational *or* irrational, never both.

Common Mistake to Avoid

A square root is not automatically irrational! $\sqrt{36} = 6$ is a whole number, because 36 is a *perfect square*. Always simplify a square root first — *then* decide which family it belongs to.

Name every subset (Natural, Whole, Integer, Rational, Irrational) that each number belongs to.

1. 9

2. -14

3. $\frac{3}{8}$

4. $\sqrt{36}$

5. $\sqrt{28}$

6. -4.75

7. 0

8. $\sqrt{5}$

2. B. Simplifying Square Roots

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If the number under the root sign is a perfect square (or a fraction of two perfect squares), the square root simplifies to an exact number.

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Simplify $\sqrt{\frac{49}{100}}$.

$$\sqrt{\frac{49}{100}} = \frac{\sqrt{49}}{\sqrt{100}} = \frac{7}{10}, \qquad \text{since } 7^2 = 49 \text{ and } 10^2 = 100.$$

9. $\sqrt{64}$

10. $\sqrt{121}$

11. $\sqrt{\frac{9}{16}}$

12. $\sqrt{\frac{25}{36}}$

3. C. Estimating Square Roots

When a number is not *a perfect square, its square root sits between two whole numbers. Find which two, then decide which one it's closer to.*

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Estimate $\sqrt{52}$ to the nearest integer.

$$6^2 = 36 \qquad 7^2 = 49 \qquad 8^2 = 64$$

Since $49 < 52 < 64$, we know $7 < \sqrt{52} < 8$. Because 52 is much closer to 49 than to 64, $\sqrt{52} \approx 7$.

Estimate each square root. Round to the nearest integer.

13. $\sqrt{40}$

Show your work:

14. $\sqrt{55}$

Show your work:

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15. $\sqrt{95}$

Show your work:

16. $\sqrt{150}$

Show your work:

4. D. Square Roots in Context

If you know the area *of a square, its side length is the square root of the area — since* $A = s^2$.

17. A framed painting is a square with an area of 50 in^2. About how long is one side, to the nearest inch?

Show your work:

18. A garden plot is a square with an area of 95 m^2. About how long is one side, to the nearest meter?

Show your work:

19. A stage floor is a square with an area of 200 ft^2. About how long is one side, to the nearest foot?

Show your work:

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5. E. Comparing Real Numbers

To compare two real numbers, rewrite both as decimals — then line them up and compare.

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Compare $3\frac{1}{4}$ and $\sqrt{11}$ using $<$, $>$, or $=$.

$$3\tfrac{1}{4} = 3.25, \qquad \sqrt{11} \approx 3.3166\ldots$$

Since $3.25 < 3.3166\ldots$, $\quad 3\frac{1}{4} < \sqrt{11}$.

Compare the numbers in each pair using $<$, $>$, or $=$.

20. $4\frac{2}{5}$ ◯ $\sqrt{19}$

Show your work:

21. -2.1 ◯ $-\frac{7}{4}$

Show your work:

22. $\frac{5}{3}$ ◯ $\sqrt{3}$

Show your work:

23. $\sqrt{70}$ ◯ 8.3

Show your work:

6. F. Ordering Real Numbers

Convert every number to a decimal first — then it's just like ordering decimals.

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24. Order these numbers from **least to greatest**:

$$3, \qquad -1, \qquad \sqrt{10}, \qquad -\frac{5}{2}, \qquad 2.6$$

Show your work:

7. G. Properties of Real Numbers

These two properties describe ways you can rearrange a sum or a product without changing the answer.

Commutative & Associative Properties

Commutative Property — you can swap the *order* of the numbers:

$$a + b = b + a \qquad\qquad a \times b = b \times a$$

Associative Property — you can move the *grouping* parentheses, as long as the order of the numbers stays the same:

$$(a + b) + c = a + (b + c) \qquad\qquad (a \times b) \times c = a \times (b \times c)$$

A quick way to tell them apart: if the *numbers switch places*, it's Commutative. If the numbers *stay in the same order* but the parentheses move, it's Associative.

Name the property shown in each equation.

25. $(4 + 9) + 6 = 4 + (9 + 6)$

26. $7 \times 12 = 12 \times 7$

27. $5 \times (3 \times 8) = (5 \times 3) \times 8$

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28. $n + 15 = 15 + n$

8. H. Writing Algebraic Expressions

Turn a word phrase into symbols by translating one piece at a time.

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Write an algebraic expression for: *4 more than three times a number* t.

"three times a number t" $= 3t$ "4 more than that" $= 3t + 4$.

29. A number k increased by 9.

30. 5 less than twice a number m.

31. The quotient of a number p and 7.

32. Renting a kayak costs an \$8 flat fee plus \$3 for every hour h you keep it.

33. The table shows the total cost of catering an event based on the number of guests. Write a rule for the total cost **in words**, and then as an algebraic expression using n for the number of guests.

Number of Guests	Total Cost
10	$(15 \cdot 10) + 40$
20	$(15 \cdot 20) + 40$
30	$(15 \cdot 30) + 40$

Show your work:

9. I. Order of Operations

Work inside parentheses first, then exponents, then multiply/divide left to right, then add/subtract left to right — PEMDAS*.*

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Simplify $30 \div (1 + 2^2)$.

$$30 \div (1 + 2^2) = 30 \div (1 + 4) = 30 \div 5 = 6.$$

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34. $36 \div (2 + 4^2)$

Show your work:

35. $5^2 - 3 \times (8 - 2)$

Show your work:

36. $(9 - 4)^2 + 3 \times 2$

Show your work:

37. $48 \div (2^3 - 2) + 5$

Show your work:

10. J. Evaluating Algebraic Expressions

Substitute the given value for each variable — then follow order of operations.

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Evaluate $2x^2 \div y$ when $x = 3$ and $y = 6$.

$$2(3)^2 \div 6 = 2(9) \div 6 = 18 \div 6 = 3.$$

38. Evaluate $4x \cdot 3 \div y$ when $x = 2$ and $y = 6$.

Show your work:

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39. Evaluate $(3a)^2 \div (b-1)$ when $a = 2$ and $b = 5$.

Show your work:

40. Evaluate $2(x+y) - z$ when $x = 4$, $y = 3$, and $z = 5$.

Show your work:

11. K. Apply: Word Problems

Read carefully, decide which idea from this worksheet applies, and show your reasoning.

41. A contractor is installing tile in a square room. Tile and labor cost \$8 per square foot, and the total job costs \$960. About how long is a side of the room, to the nearest foot?

Show your work:

42. If you lean a ladder against a wall, a safe length for the ladder is $\sqrt{x^2 + (3x)^2}$ feet, where x is how far the base of the ladder sits from the wall. Find the ladder length, to the nearest tenth of a foot, when the base is 6 feet from the wall.

Show your work:

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43. A rafting company charges \$48 per adult, \$27 per child, and \$6 per hour for equipment rental. Write an expression for the total cost of a trip for 2 adults and 3 children who rent equipment for h hours. Then find the total cost if the trip lasts 4 hours.

Show your work:

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Detailed Answer Key

Full solutions and reasoning, in order.

Section A — Classifying Real Numbers.

1. 9: Natural, Whole, Integer, Rational.
2. -14: Integer, Rational (not Natural or Whole — it's negative).
3. $\frac{3}{8}$: Rational only.
4. $\sqrt{36} = 6$: Natural, Whole, Integer, Rational (it's a perfect square!).
5. $\sqrt{28} \approx 5.2915\ldots$: Irrational ($28$ is not a perfect square, so the decimal never ends or repeats).
6. -4.75: Rational (a terminating decimal, and $-4.75 = -\frac{19}{4}$).
7. 0: Whole, Integer, Rational (not Natural — the natural numbers start at 1).
8. $\sqrt{5} \approx 2.2360\ldots$: Irrational ($5$ is not a perfect square).

Section B — Simplifying Square Roots.

9. $\sqrt{64} = 8$, since $8^2 = 64$.
10. $\sqrt{121} = 11$, since $11^2 = 121$.
11. $\sqrt{\frac{9}{16}} = \frac{\sqrt{9}}{\sqrt{16}} = \frac{3}{4}$.
12. $\sqrt{\frac{25}{36}} = \frac{\sqrt{25}}{\sqrt{36}} = \frac{5}{6}$.

Section C — Estimating Square Roots.

13. $\sqrt{40}$: since $36 < 40 < 49$, $6 < \sqrt{40} < 7$; 40 is close to 36, so $\sqrt{40} \approx 6$ ($\sqrt{40} \approx 6.32$).
14. $\sqrt{55}$: since $49 < 55 < 64$, $7 < \sqrt{55} < 8$; 55 is closer to 49, so $\sqrt{55} \approx 7$ ($\sqrt{55} \approx 7.42$).
15. $\sqrt{95}$: since $81 < 95 < 100$, $9 < \sqrt{95} < 10$; 95 is much closer to 100, so $\sqrt{95} \approx 10$ ($\sqrt{95} \approx 9.75$).
16. $\sqrt{150}$: since $144 < 150 < 169$, $12 < \sqrt{150} < 13$; 150 is much closer to 144, so $\sqrt{150} \approx 12$ ($\sqrt{150} \approx 12.25$).

Section D — Square Roots in Context.

17. $A = s^2 = 50 \Rightarrow s = \sqrt{50} \approx 7.07$, so the side is about $\mathbf{7}$ inches.
18. $A = s^2 = 95 \Rightarrow s = \sqrt{95} \approx 9.75$, so the side is about $\mathbf{10}$ meters.
19. $A = s^2 = 200 \Rightarrow s = \sqrt{200} \approx 14.14$, so the side is about $\mathbf{14}$ feet.

Section E — Comparing Real Numbers.

20. $4\frac{2}{5} = 4.4$ and $\sqrt{19} \approx 4.359$, so $4\frac{2}{5} > \sqrt{19}$.
21. -2.1 and $-\frac{7}{4} = -1.75$; since -2.1 is farther left on the number line, $-2.1 < -\frac{7}{4}$.
22. $\frac{5}{3} \approx 1.667$ and $\sqrt{3} \approx 1.732$, so $\frac{5}{3} < \sqrt{3}$.
23. $\sqrt{70} \approx 8.367$ and 8.3, so $\sqrt{70} > 8.3$.

Section F — Ordering Real Numbers.

24. Converting: $3 = 3.000$, $-1 = -1.000$, $\sqrt{10} \approx 3.162$, $-\frac{5}{2} = -2.5$, $2.6 = 2.600$.

From least to greatest: $\mathbf{-\frac{5}{2}}$, $\mathbf{-1}$, $\mathbf{2.6}$, $\mathbf{3}$, $\mathbf{\sqrt{10}}$.

Section G — Properties of Real Numbers.

25. $(4 + 9) + 6 = 4 + (9 + 6)$: the order of the numbers doesn't change, only the grouping — **Associative Property of Addition**.
26. $7 \times 12 = 12 \times 7$: the two numbers switch places — **Commutative Property of Multiplication**.

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27. $5 \times (3 \times 8) = (5 \times 3) \times 8$: the order stays the same, only the grouping moves — **Associative Property of Multiplication**.
28. $n + 15 = 15 + n$: the two terms switch places — **Commutative Property of Addition**.

Section H — Writing Algebraic Expressions.

29. $k + 9$.
30. $2m - 5$.
31. $\frac{p}{7}$ (or $p \div 7$).
32. $8 + 3h$.
33. Each cost is \$15 times the number of guests, plus a \$40 fee. In words: "the total cost is \$15 per guest plus a \$40 flat fee." As an expression with n guests: $\mathbf{15n + 40}$.

Section I — Order of Operations.

34. $36 \div (2 + 4^2) = 36 \div (2 + 16) = 36 \div 18 = \mathbf{2}$.
35. $5^2 - 3 \times (8 - 2) = 25 - 3 \times 6 = 25 - 18 = \mathbf{7}$.
36. $(9 - 4)^2 + 3 \times 2 = 5^2 + 6 = 25 + 6 = \mathbf{31}$.
37. $48 \div (2^3 - 2) + 5 = 48 \div (8 - 2) + 5 = 48 \div 6 + 5 = 8 + 5 = \mathbf{13}$.

Section J — Evaluating Algebraic Expressions.

38. $4(2) \cdot 3 \div 6 = 24 \div 6 = \mathbf{4}$.
39. $(3 \cdot 2)^2 \div (5 - 1) = 6^2 \div 4 = 36 \div 4 = \mathbf{9}$.
40. $2(4 + 3) - 5 = 2(7) - 5 = 14 - 5 = \mathbf{9}$.

Section K — Apply: Word Problems.

41. Area $= \$960 \div \$8$ per ft^2 $= 120$ ft^2. Side $= \sqrt{120} \approx 10.95$, so the room is about $\mathbf{11}$ feet on a side.
42. Length $= \sqrt{6^2 + (3 \cdot 6)^2} = \sqrt{36 + 324} = \sqrt{360} \approx 18.9736\ldots$, which rounds to $\mathbf{19.0}$ feet (nearest tenth).
43. Cost $= 2(48) + 3(27) + 6h = 96 + 81 + 6h = \mathbf{177 + 6h}$. At $h = 4$: $177 + 6(4) = 177 + 24 = \$\mathbf{201}$.

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